Higher National Diploma in Information Technology
Second Year, First Semester Examination – 2022
HNDIT3072 – Statistics for IT

Instructions for Candidates:
Answer any five (05) questions.
Every question carries 20 marks
You are allowed to use non-programable calculator.

No. of questions : 06
No. of pages : 05
Time : Three (03) hours

Question 01.

(i). Mention the importance of probability. **(4 marks)**

(ii). Suppose you toss a die once and consider the following events;

A – observe an odd number.

B – observe a number greater than 2.

C – observe 6.

D – observe 3.

(a). Identify the experiment you have done as mentioned in the above. **(1 marks)**

(b). Find an event of that experiment you mentioned in part (a). **(1 marks)**

(c). Write the sample space of the experiment in part (a). **(1 marks)**

(d). Find two (02) mutually exclusive events from the provided details in the question. **(1 marks)**

(iii). There are 120 students in a class. If the probability of selected student being a boy is 0.7. Moreover, the number of students (both boys and girls) with blue-eyes in the class is 40.

(a). What is the probability of the selected student being a girl student? Show the answer using probability rule. **(4 marks)**

(b). What is the probability of the selected student being a non-blue-eye student? **(4 marks)**

(c). If the probability of selected student being a boy with blue-eyes is 0.167, find the probability of the selected student with blue-eyes given that the selected student is a boy. (Hint: probability of the selected student being a boy is 0.7). **(4 marks)**

[20 Marks]

Question 02

(i). Define the term" a Probability Distribution". **(2 marks)**

(ii). Write two (02) key attributes of Binomial probability distribution. **(2 marks)**

(iii). Consider the discrete random variable probability distribution of tossing three coins together given below.

No. of heads (x)	**0**	**1**	**2**	**3**
P(X)	0.125	0.375	a	0.125

For the above find;

(a). the value a. **(4 marks)**

(b). the expected value. **(4 marks)**

(iv). A biased coin has a probability of 0.4 of getting a head. Assume the coin is tossed for 3 times. Using Binomial probability distribution, find;

(a). the probability of getting exactly 3 heads. **(4 marks)**

(b). expected value and variance **(4 marks)**

[20 Marks]

Question 03

(i). List two (02) importance and two (02) limitation of Statistics. **(4 marks)**

(ii). Write 04 of probabilistic sampling methods and describe them in brief. **(4 marks)**

(iii). Sample of 30 students is showed no. of questions they have done as follows;

5	10	50	40	12	42	12	50	45	20
30	8	25	28	49	41	36	26	13	44
50	34	22	50	16	1	50	50	30	22

For the above;

(a). Prepare a frequency distribution by selecting class width as 10 and by taking the first class as 1-10. **(4 marks)**

(b). Construct a histogram and a frequency polygon. **(4 marks)**

(c). Draw the less than ogive. **(4 marks)**

[20 Marks]

Question 04

(i). What is the importance of measures of central tendency of a data set? Mention 02 of common measures of central tendency in Statistics. **(4 marks)**

(ii). No of leaves taken by 10 employees are listed here.

5, 6, 8, 4, 5, 6, 1, 3, 8, 2

For the above find:

(a). The average no of leaves taken by an employee. **(1 marks)**

(b). The median value of taken leaves. **(1 marks)**

(c). The mode **(1 marks)**

(d). How do you interpret/ explain the answer received as mode (in part c)? **(1 marks)**

(iii). Marks scored by 100 students for Statistics for IT assignment are as follows;

Marks	No. of Students
51 - 55	10
56 - 60	25
61 - 65	17
66 - 70	20
71 - 75	18
76 - 80	10

For the above assignment marks find the followings;

(a). The average mark. **(4 marks)**

(b). The expected mark in the middle. **(4 marks)**

(c). The most frequent mark. **(4 marks)**

[20 Marks]

Question 05.

(i). What does is mean by the measures of dispersion in statistics? Write 02 of measures of dispersion. **(4 marks)**

(ii). "Mean is not a good statistic measure, if outlies are there in the data set". Do you agree with the statement? Justify your answer. **(4 marks)**

(iii). Attendance (in days) of 7 students in the English class is as follows;

15 20 40 3 32 22 51

For the above marks find the followings;

(a). First and third quartiles. **(2 marks)**

(b). Inter quartile range. **(2 marks)**

(c). The sample variance. **(4 marks)**

(d). The standard deviation. **(2 marks)**

(e). Coefficient of variance (CV). **(2 marks)**

[20 Marks]

Question 06.

(i). Define the following terms related to the correlation;

(a). Positive correlation **(1 marks)**

(b). Negative correlation **(1 marks)**

(c). Independent variable **(1 marks)**

(d). Dependent variable **(1 marks)**

(ii). Mr. Perera, the HRM manager, did a study on job satisfaction and life satisfaction with his office staff. Participants completed a measure on how job satisfaction they were feeling (on a 1 to 10 scale) and a measure of how satisfied they felt with their lives (measured on a 1 to 20 scale). The table below indicates the participants' scores.

Employee	Job Satisfaction (x)	Life Satisfaction (y)
1	3	6
2	2	2
3	4	9
4	5	10
5	6	15

Considering the provided information;

(a). draw a scatter plot and find the type of linear correlation. **(4 marks)**

(b). find the correlation coefficient. **(4 marks)**

(iii). Consider the following information.

Production Level (x) (in units)	Profit (y) (Rs. "000")
10	14
20	25
30	40
40	60
50	85
60	100

For the above information find;

(a). regression line using least square regression line y on x. **(6 marks)**

(b). the expected profit when the production is 100 units. **(2 marks)**

[20 Marks]